

The image features the text "DL for ENG" in a bold, 3D, gold-colored font. The letters are floating on a vast, blue ocean with visible ripples and small waves. In the background, there are rolling blue hills or mountains under a sky filled with soft, grey clouds. The overall scene has a serene, digital aesthetic.

DL for ENG

LECTURE 9.1

Laminar Flow Prediction

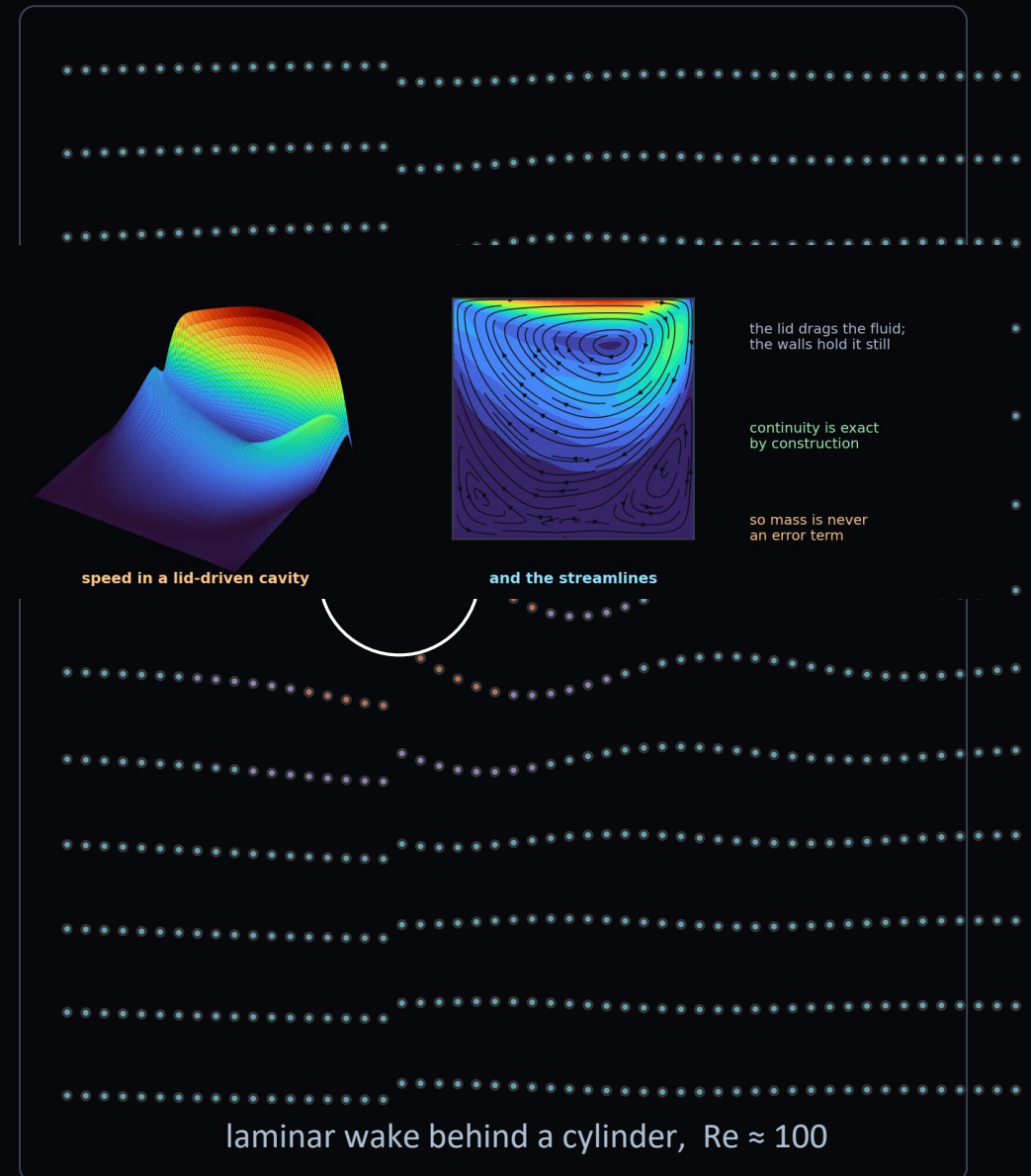
From Burgers to incompressible Navier–Stokes: the first vector-valued, coupled, genuinely nonlinear problem in the course

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Raissi, Perdikaris & Karniadakis (2019) · Liu (2025) for the machinery



What You Will Be Able To Do

THE GOALS, IN PLAIN WORDS

- **write** — the three residuals, and count the derivatives each one needs
- **explain** — what the Reynolds number sets, both physically and for the method
- **use** — a stream function, and say what it buys and what it costs
- **handle** — a pressure field that is only determined up to a constant
- **place** — collocation points where flow is actually difficult
- **state** — the case for a PINN over a finite-volume code, narrowly

WHAT TODAY ASSUMES

from L7.2	Burgers, and the convection nonlinearity
from L8	scaling, hard enforcement, verification
from L8.2	time as a coordinate

FOUR THINGS CHANGE AT ONCE

Three coupled fields, a nonlinear term, a constraint with no time derivative, and a pressure that is not physically determined.

AND ONE RESULT WORTH THE HOUR

Pressure recovered from velocity alone, with the viscosity, from noisy data.

Four Things Change at Once

	L8 — HEAT	L9 — FLOW
Unknown	<i>one scalar T</i>	<i>three fields: u, v, p</i>
Equations	<i>one PDE</i>	<i>two momentum + one constraint</i>
Nonlinearity	<i>none (or mild)</i>	<i>convection $(u \cdot \nabla)u$</i>
Coupling	—	<i>pressure couples both momenta</i>
Constraint	—	<i>incompressibility, no time derivative</i>
Difficulty knob	<i>Bi, Fo</i>	<i>Reynolds number</i>

The machinery is still the machinery. What is new is that the network now has three outputs, the residual has three components, and one of them is a constraint rather than an evolution equation.

Incompressible Navier–Stokes

MASS CONSERVATION — A CONSTRAINT, NOT AN EVOLUTION EQUATION

$$\nabla \cdot \mathbf{u} = \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0$$

MOMENTUM CONSERVATION

$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla) \mathbf{u} = -\frac{1}{\rho} \nabla p + \nu \nabla^2 \mathbf{u}$$

- four terms in the momentum equation, each with a physical meaning
- the convection term is the source of every difficulty in this lecture
- it is nonlinear, and it couples the components
- **continuity has no time derivative and no source** — it constrains rather than evolves
- delete convection and you have Stokes flow, which behaves like the heat problems

The Residual, Component by Component

X-MOMENTUM

$$u_{,t} + u u_{,x} + v u_{,y} = -\frac{1}{\rho} p_{,x} + \nu(u_{,xx} + u_{,yy})$$

Y-MOMENTUM

$$v_{,t} + u v_{,x} + v v_{,y} = -\frac{1}{\rho} p_{,y} + \nu(v_{,xx} + v_{,yy})$$

CONTINUITY

$$\nabla \cdot \mathbf{u} = \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0$$

- three residuals, all from the same graph, all at the same points
- **count the derivatives** — one in time, two first in space, two second
- **autograd handles the nonlinearity exactly** — no linearisation, no outer iteration

The Reynolds Number Is the Difficulty Knob

DEFINITION

$$Re = \frac{UL}{\nu} = \frac{\text{inertia}}{\text{viscosity}}$$

NON-DIMENSIONAL FORM — RE IS THE ONLY PARAMETER

$$\mathbf{u}_{,t}^* + (\mathbf{u}^* \cdot \nabla) \mathbf{u}^* = -\nabla p^* + \frac{1}{Re} \nabla^2 \mathbf{u}^*$$

- non-dimensionalise and every material property collapses into one number
- two flows with the same Reynolds number behave identically
- and it measures how hard the problem is for a PINN
- low Re is smooth; high Re is thin layers and sharp gradients
- the practical ceiling on the raw equations is a few hundred

REGIMES PAST A CYLINDER

$Re < 5$ creeping, attached

$5 - 40$ steady twin vortices

$40 - 200$ vortex shedding

$200 - 10^5$ wake turbulent

$> 10^5$ fully turbulent

A plain PINN is comfortable in the first three bands and increasingly unreliable beyond them.

2-D Burgers: Navier–Stokes Without the Pressure

X-COMPONENT

$$u_{,t} + u u_{,x} + v u_{,y} = \nu(u_{,xx} + u_{,yy})$$

Y-COMPONENT

$$v_{,t} + u v_{,x} + v v_{,y} = \nu(v_{,xx} + v_{,yy})$$

- take Navier–Stokes, delete the pressure gradient, drop incompressibility
- what is left is 2-D Burgers
- it keeps the convection nonlinearity and the vector coupling
- **so it is the stepping stone** — and Ex_9.1 starts here
- if your Burgers solver works, adding pressure is incremental

WHAT EACH MODEL KEEPS

	BURGERS	N–S
convection $(\mathbf{u} \cdot \nabla) \mathbf{u}$	yes	yes
viscous diffusion	yes	yes
vector coupling	yes	yes
pressure gradient	no	yes
incompressibility	no	yes

the two new difficulties arrive first, alone

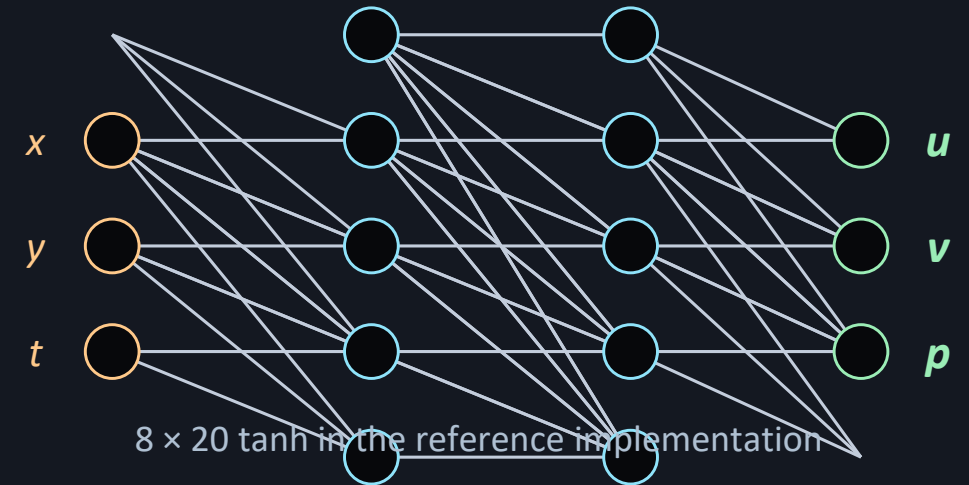
One Network, Three Outputs

THE ANSATZ

$$(u, v, p) = \mathcal{N}_{\hat{w}}(x, y, t)$$

- one network with three outputs, not three networks
- the fields share hidden representations, which is the point
- pressure enters only through its gradient, so its absolute level is never needed
- depth matters more here than in the heat problems
- **tanh throughout** — the residual needs second derivatives

SHARED TRUNK, THREE HEADS



Three outputs, three residuals — but only two of them are evolution equations. The third constrains the first two.

Three Ways to Impose Incompressibility

PENALTY

Add $\|\nabla \cdot \mathbf{u}\|^2$ as another loss term.

Simplest to code.
Only approximately satisfied,
and it competes with the
momentum residuals for the
optimiser's attention.

STREAM FUNCTION

Predict ψ , take u and v from its
derivatives.

Divergence-free by construction,
to machine precision.
One fewer output and one fewer
loss term.
2-D only. Raissi's choice.

PROJECTION

Correct the velocity onto a
divergence-free space.

Standard in classical CFD.
Awkward inside a PINN — it
needs a Poisson solve, which
is exactly what we were
trying to avoid.

STREAM FUNCTION DEFINITION

$$u = \frac{\partial \psi}{\partial y}, \quad v = -\frac{\partial \psi}{\partial x}$$

WHY IT WORKS: MIXED PARTIALS COMMUTE

$$\nabla \cdot \mathbf{u} = \psi_{,yx} - \psi_{,xy} \equiv 0$$

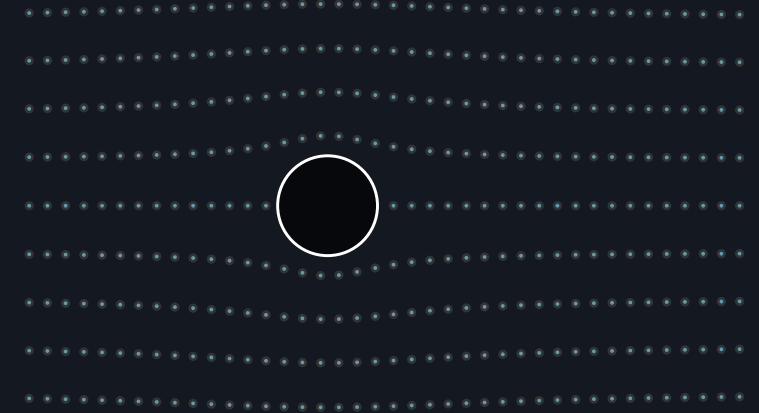
The Stream Function in Practice

VELOCITY FROM Ψ

$$u = \frac{\partial \psi}{\partial y}, \quad v = -\frac{\partial \psi}{\partial x}$$

- the network predicts the stream function and the pressure
- velocities come from differentiating it
- continuity is then satisfied structurally, and its loss term disappears
- the cost is third derivatives in the momentum residual
- **and the limitation is real** — a stream function exists only in 2-D or axisymmetric flow

CONTOURS OF Ψ ARE STREAMLINES



$\nabla \cdot \mathbf{u} = 0$ holds exactly, at every point, always

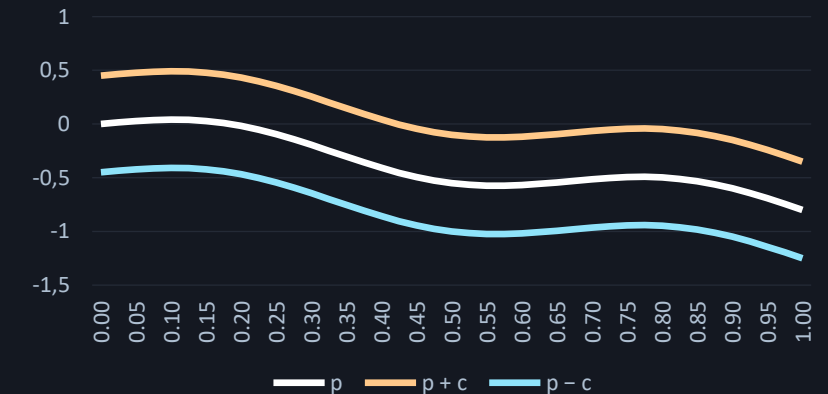
Pressure Is Only Determined Up to a Constant

THE GAUGE FREEDOM

$$p \rightarrow p + c \Rightarrow \text{same velocity field}$$

- in incompressible flow the pressure has no thermodynamic meaning
- it appears only through its gradient, so any constant may be added
- the same degeneracy as the pure-Neumann heat problem in L8.1
- so never compare absolute pressures between runs
- **fix it by prescribing zero at one point** — it costs nothing

SAME FLOW, THREE PRESSURE FIELDS



identical gradients — identical velocity

Boundary Conditions for a Flow Problem

NO-SLIP WALL

$$\mathbf{u} = 0 \quad \text{on the wall}$$

Both velocity components vanish on a solid surface. The single most important condition in viscous flow — it is what creates boundary layers, drag and wakes.

INLET

$$u = U(y), \quad v = 0 \quad \text{at the inlet}$$

A prescribed velocity profile. Uniform is simplest; fully developed Poiseuille is more realistic for a pipe and avoids a spurious entrance region.

OUTLET

$$p = 0, \quad \frac{\partial u}{\partial n} = 0 \quad \text{at the outlet}$$

Pressure fixed and velocity gradients zero — the outflow leaves without reflecting. This is also where you fix the pressure gauge.

$$u(y) = \frac{3}{2}U_{avg}\left[1 - \left(\frac{2y}{H}\right)^2\right]$$

The Composite Loss for Flow

FIVE TERMS, AT MINIMUM

$$\mathcal{L} = \mathcal{L}_u + \mathcal{L}_v + w_c \mathcal{L}_{cont} + \mathcal{L}_{BC} + \mathcal{L}_{IC}$$

- two momentum residuals, one continuity residual, conditions on every surface
- **the continuity term is the one to watch** — different units from the momentum terms
- a stream function removes it entirely
- which is the strongest practical argument for that formulation
- and the no-slip condition deserves its own weight

TERM BY TERM

$\mathcal{L}_u, \mathcal{L}_v$	momentum	<i>always</i>
$\mathcal{L}^{cont}$	continuity	<i>removable by ψ</i>
$\mathcal{L}^{bc}$	no-slip, inlet, outlet	<i>weight the wall</i>
$\mathcal{L}_i^c$	initial field	<i>unsteady only</i>
$\mathcal{L}^{data}$	measurements	<i>inverse only</i>

Non-Dimensionalisation Is Not Optional Here

CHANGE OF VARIABLES

$$u^* = \frac{u}{U}, \quad x^* = \frac{x}{L}, \quad p^* = \frac{p}{\rho U^2}$$

- air has a viscosity around 1.5e-5, water around 1e-6 — raw numbers span orders
- scaled, all of that becomes one over the Reynolds number
- one number replaces four, and the fields become order one
- note the pressure scaling — by density times velocity squared, not by anything viscous
- train scaled, present unscaled, and state which you are showing

WATER IN A SMALL PIPE

QUANTITY	PHYSICAL	SCALED
diameter D	0.05 m	1
velocity U	0.2 m/s	1
viscosity ν	1.0e-6	1/Re
Reynolds	—	10 000
pressure	$\rho U^2 = 40 \text{ Pa}$	1

Re = 10 000 in a real pipe — turbulent, and beyond a plain PINN. That gap is L9.2.

Where the Points Must Go

- flow is not uniformly difficult
- almost everything happens in the boundary layer and in the wake
- the boundary layer thins as the square root of Re
- so cluster points near walls and along the wake centreline
- residual-adaptive refinement works well here and is worth implementing

UNIFORM VS. GRADED SAMPLING



uniform — boundary layer unresolved

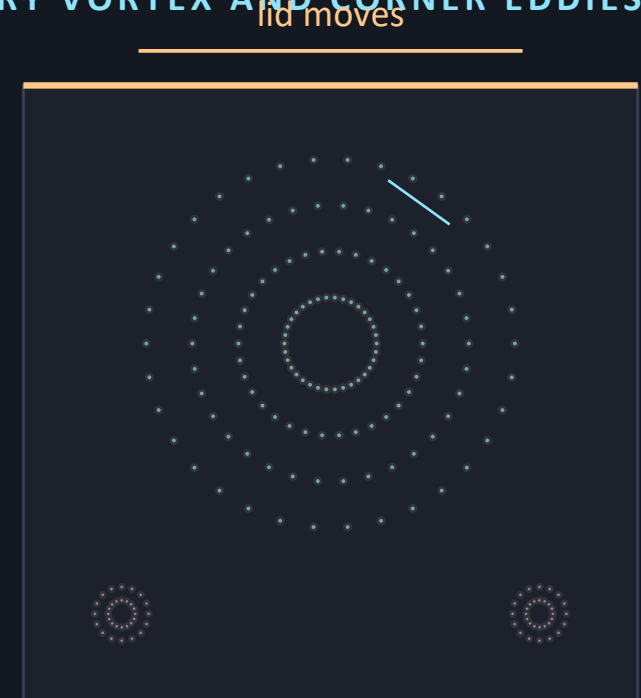


graded — points where the physics is

The Lid-Driven Cavity

- a square box, three fixed walls, and a lid sliding at constant speed
- **no inlet, no outlet, no obstacle** — the geometry could not be simpler
- and tabulated reference velocities exist across the centrelines
- it has one deliberate nastiness: the lid corners are singular
- **start here** — if the cavity fails, nothing downstream will work

PRIMARY VORTEX AND CORNER EDDIES



counter-rotating corner eddies — a real test of resolution

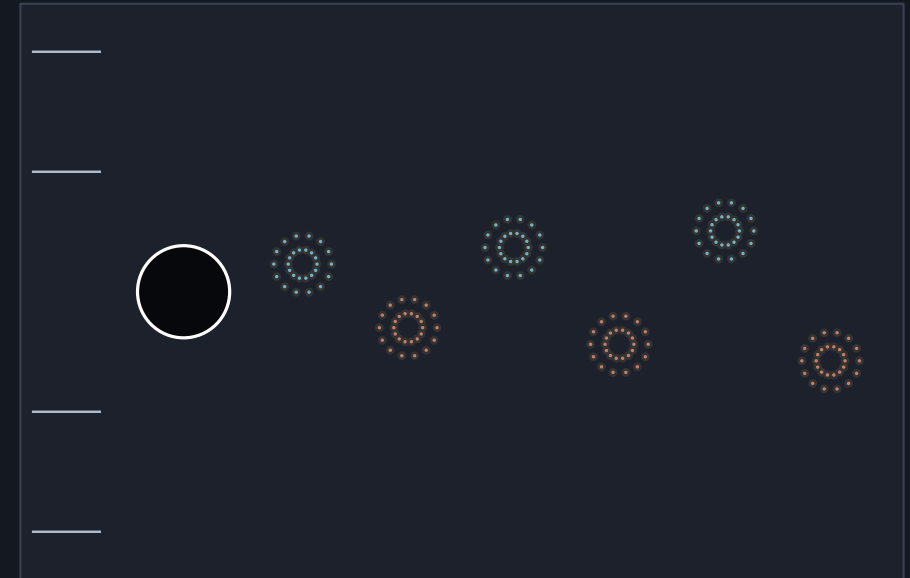
Flow Past a Cylinder at $Re = 100$

SHEDDING FREQUENCY

$$St = \frac{fD}{U} \approx 0.2 \quad (Re \sim 100)$$

- the canonical unsteady benchmark, and the one Raissi uses
- at $Re = 100$ the wake sheds periodically
- **unsteady but not chaotic** — one dominant frequency
- the Strouhal number is the verification target, not the field
- a model that reproduces the field but not the frequency has not solved the problem

THE KÁRMÁN VORTEX STREET



alternating vortices, shed at a single frequency

How to Know a Flow Solution Is Right

KOVASZNAY FLOW — AN EXACT STEADY SOLUTION OF NAVIER-STOKES

$$u = 1 - e^{\lambda x} \cos(2\pi y), \quad \lambda = \frac{Re}{2} - \sqrt{\frac{Re^2}{4} + 4\pi^2}$$

EXACT SOLUTIONS

Kovasznay flow above, and Taylor–Green for the unsteady case. Both satisfy the full nonlinear equations, so they test everything at once.

REFERENCE TABLES

Ghia’s cavity centreline velocities at $Re = 100, 400, 1000$. Compare pointwise; disagreement beyond a few percent is a bug.

INTEGRAL QUANTITIES

Drag and lift coefficients, and the Strouhal number for shedding. These are what an engineer reports, and they are sensitive to the boundary layer.

DIVERGENCE CHECK

Evaluate $\nabla \cdot \mathbf{u}$ on a dense grid. With a stream function it must be at machine precision; with a penalty it tells you how well the constraint was enforced.

A pretty streamline plot proves nothing. Every one of these four is a number you can be wrong about.

Recovering Pressure From Velocity Alone

VISCOSITY AS A TRAINABLE PARAMETER

$$\nu \text{ trainable}, \quad \mathcal{L} = \mathcal{L}_{PDE} + \mathcal{L}_{data}$$

- this is the result that made PINNs interesting to the fluids community
- **velocity can be measured** — by PIV, by ultrasound, by dye
- pressure inside a flow generally cannot
- train on scattered velocity with the residual in the loss, and pressure follows
- the same run recovers the viscosity to about one per cent from noisy data

WHAT GOES IN, WHAT COMES OUT

MEASURED

u, v at scattered points

IMPOSED

the N-S residual

RECOVERED

p, everywhere

RECOVERED

pressure is never measured, and never needs to be
v, to ~1%

Failure Modes in Laminar Flow

Continuity ignored

The divergence residual is orders of magnitude smaller than the momentum ones and gets outvoted.

Symptom: mass not conserved, streamlines that start or stop. Fix: stream function, or weight it.

Boundary layer unresolved

Too few points within $\delta \sim \text{Re}^{-1/2}$ of the wall for no-slip to bite.

Symptom: drag far too low, wake too weak. Fix: grade the sampling towards walls.

Trivial uniform flow

A constant velocity field satisfies the momentum equations and continuity exactly.

Symptom: the obstacle appears to have no effect. Fix: check the no-slip loss separately.

Pressure drift

Gauge freedom left unfixed, so absolute pressure differs run to run.

Symptom: irreproducible pressures. Fix: pin p at the outlet.

Re too high

Convection dominates, gradients steepen, and the smooth ansatz cannot follow.

Symptom: training stalls at a smooth, wrong field. This is the honest limit — see L9.2.

The third is the one students miss: a uniform field is a perfectly valid solution of the equations — just not of the boundary conditions.

PINN or CFD for a Laminar Flow?

USE CFD

A single forward solve on fixed geometry with known properties.
Anything above $Re \sim 10^3$.
You need a validated answer with grid-convergence evidence.
Forty years of verification behind it.

USE A PINN

Reconstructing a field from sparse measurements — the PIV problem.
Recovering pressure or viscosity that cannot be measured.
Geometry or parameters as inputs, so one training serves a sweep.

- for a laminar forward solve, a finite-volume code wins
- faster, more accurate, and far better verified

Judge the method by what it makes possible, not by whether it outperforms a mature solver at the solver's own job.

Questions

TEN TO TEST WHETHER TODAY LANDED

- which term causes the difficulty, and why?
- what does continuity constrain, given it has no time derivative?
- what does the Reynolds number tell you about your chances?
- what does a stream function guarantee, and what does it cost?
- why is the pressure only determined up to a constant?
- where do the collocation points belong, and why not uniformly?

AND FOUR MORE

- seven** why the cavity before any obstacle problem?
- eight** what verifies a cylinder result, if not the field?
- nine** how is pressure recovered if never measured?
- ten** when would you use a finite-volume code instead?

THE ONE WITH NO SETTLED ANSWER

The lid corners are singular. What should the network do there?

BEFORE L9.2

Ex_09.1 notebooks 01 to 03. Bring the cavity centreline profile.

Ex_9.1 — 2-D Burgers with an Interactive Interface

TASKS

- 1 Solve 2-D Burgers for the coupled field (u, v) on the unit square with a known exact solution.
- 2 Use the control panel to vary the collocation count and record how the error responds.
- 3 Sweep the Reynolds number upward and find where your model stops converging.
- 4 Compare a shared-trunk network against two separate networks at equal parameter count.
- 5 Generate the automatic report and interpret every figure in it.

QUESTIONS — AND WHERE THIS DECK ANSWERS THEM

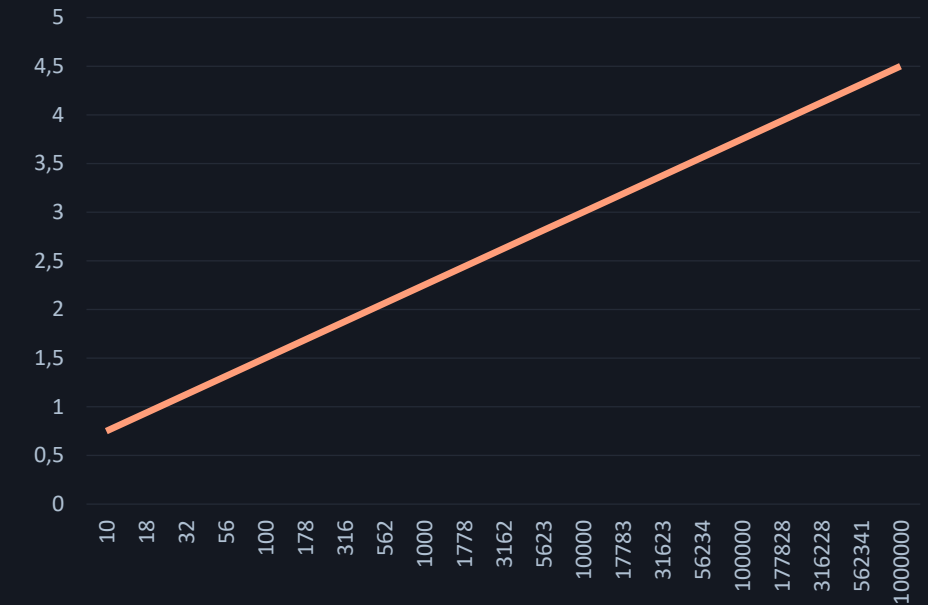
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|--|----------|
| Why is convection the source of every difficulty? | sl. 3, 6 |
| Why does Re measure difficulty as well as physics? | sl. 5 |
| Why does a stream function remove a loss term? | sl. 8, 9 |
| Why is pressure only determined up to a constant? | sl. 10 |
| Why does uniform sampling under-resolve a wall? | sl. 14 |

The interface lets you change parameters without touching code — but every question above needs an explanation, not a screenshot.

What L9.2 Has to Deal With

- everything here assumed the flow is smooth, and below a few hundred it is
- above the transition it becomes chaotic and multiscale
- a PINN on the raw equations does not converge to the turbulent solution
- it converges to something smooth, and wrong
- **so L9.2 changes the question** — model turbulence rather than resolve it

SCALE RANGE GROWS WITH RE



The ratio of largest to smallest eddy grows as $Re^{3/4}$. At $Re = 10^4$ that is a factor of a thousand in each direction — which is why turbulence is expensive for every method, not just this one.

Key Takeaways

- 1 Convection is the difficulty** Delete it and flow behaves like heat. Keep it and you have nonlinearity, coupling and a Reynolds number that measures how hard the problem is.
- 2 Continuity is a constraint** It has no time derivative and no source. The pressure is the multiplier that enforces it — which is why pressure has no absolute value.
- 3 Hard enforcement again** A stream function makes $\nabla \cdot \mathbf{u} = 0$ exact and deletes a loss term. The same argument as Dirichlet in L7, initial conditions in L8, orthogonality in L7.2.
- 4 Sample where the physics is** Boundary layers thin as Re grows. Uniform sampling silently fails to resolve them, and drag is the first thing to go wrong.
- 5 The win is assimilation** For a forward laminar solve, use CFD. For pressure from velocity, or viscosity from noisy data, there is no classical equivalent.

Next — L9.2: Turbulent Flow, where the question changes from solving to modelling.